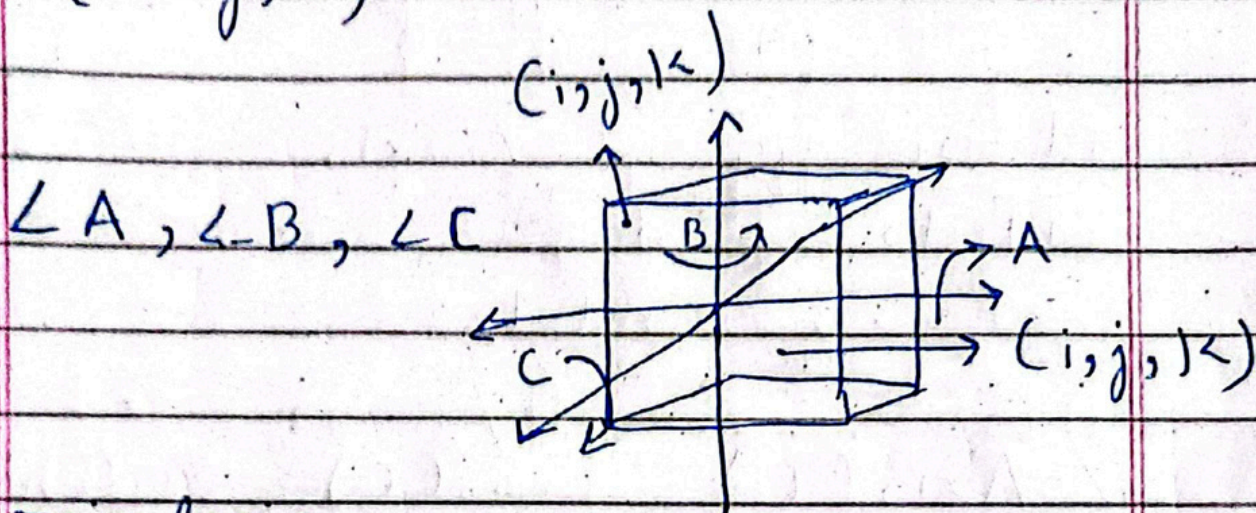


## Rotating 3D cube:-

- ) cubes existing in 3D so to represent its vertices are  $(x, y, z) \rightarrow$



main logic

↳ we are not rotating surfaces but the points on its surface

this rotation will be done by three angles  $A$  (respective to  $x$ ),  $B$  (respective to  $y$ ) and  $C$  (respective to  $z$ ).

$$X\text{-axis}(A) \Rightarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos(A) & -\sin(A) \\ 0 & \sin(A) & \cos(A) \end{bmatrix}$$

rotation around  $x$ -axis



effects x stays same  
but y & z rotate

$$Y\text{-Axis}(B) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos(B) & \sin(B) \\ 0 & -\sin(B) & \cos(B) \end{bmatrix}$$

rotation around Y-axis

effects y stays same  
but x & z rotates

Scanned with CamScanner

$$Z\text{-Axis}(C) = \begin{bmatrix} \cos(C) & -\sin(C) & 0 \\ \sin(C) & \cos(C) & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

z stays same  
but x & y rotates

→ To rotate a point we will multiply each coordinate

$$(x, y, z) = R_x(A) \cdot X \cdot R_y(B) \cdot R_z(C) \cdot (i, j, k)$$



Step 1:

$$(R_x(A) \times R_y(B)) = \begin{bmatrix} \cos(B) & 0 & \sin(B) \\ \sin(A)\sin(B) & \cos(A) & \sin(A)\cos(B) \\ -\cos(A)\sin(B) & -\sin(A) & \cos(A)\cos(B) \end{bmatrix}$$

$$\times R_z(C) =$$

col-1

$$\begin{bmatrix} \cos(B)\cos(C) \\ \cos(A)\sin(C) + \sin(A)\sin(B)\cos(C) \\ \sin(A)\sin(C) - \cos(A)\sin(B)\cos(C) \end{bmatrix}$$

col-2

$$\begin{bmatrix} -\cos(B)\sin(C) \\ \cos(A)\cos(C) - \sin(A)\sin(B) \\ \cos(A)\cos(C) + \sin(B)\sin(A)\sin(C) \end{bmatrix}$$

col-3

$$\begin{bmatrix} \sin(B) \\ -\sin(A)\cos(B) \\ \cos(A)\cos(B) \end{bmatrix}$$

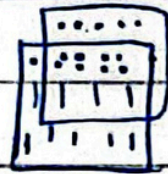
then by  $i, j, k \rightarrow \begin{matrix} & i \\ k & \nearrow \\ & j \end{matrix}$



We will be done rotating  
each point surfaces

→ using Z-buffering

↳  
for depth  
perception

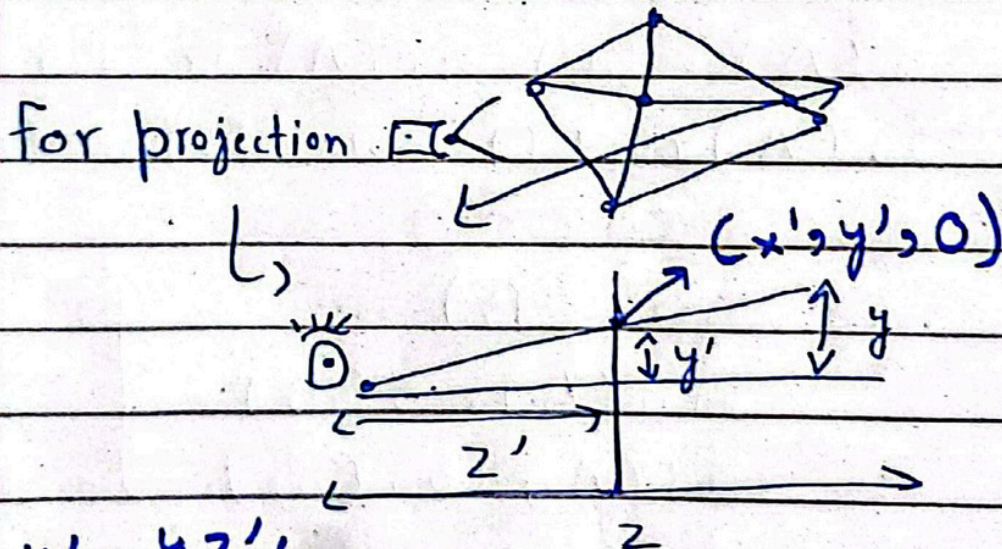


as 3D points project onto same  
screen.

for

this  $o_o z = 1f/z$  is used

$x_p$  &  $y_p$  are project which  
are calculated



$$y' = y \frac{z'}{z} \quad , \quad y\text{-projection}$$

$$x' = \frac{x z'}{z} \quad , \quad x\text{-projection}$$